

No. of Printed Pages : 4
Roll No.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the working and constructional details of a floating holder type biogas plant with the help of suitable sketch.
- Q.37 Explain the working and construction of flat plate box solar cooker with the help of suitable sketch. Also write its advantages and disadvantages.
- Q.38 Discuss about importance and scope of energy conservation. Give suggestion to minimize the wastage of energy.

4th Semester/ Agri

**Subject : Renewable Sources of Energy/
Non-Conv. Engy. Resources**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not a non-renewable energy resource?
- a) Biogas b) Coal
- c) Diesel d) Solar
- Q.2 The proportion of wet cow dung and water feed to the digester is
- a) 1:1 b) 1:2
- c) 1:3 d) 1:4
- Q.3 Tidal energy is direct associated with
- a) Sun b) Sea
- c) Wind d) All of these
- Q.4 Which of these energy resources is/are widely used in industries?
- a) Crop Residue b) Biogas
- c) Wood d) Coal and Gasoline
- Q.5 Anemometer is used to measure the is called
- a) Sun radiation b) Wind temperature
- c) Wind speed d) Sun direction

- Q.6 Solar energy can be converted into electric energy by using of
 a) Generator b) Aluminum cell
 c) Photovoltaic cell d) All of these
- Q.7 Which one is not related with wind energy?
 a) HAWT b) Darious motor
 c) Savonious rotor d) Photo-voltaic
- Q.8 Important element of solar cell is
 a) Silicion b) Silver
 c) Aluminum d) Argon
- Q.9 Which of the following biochemical conversion process is performed by microganisms?
 a) Anaerobic b) Composting
 c) Fermentation d) All of these
- Q.10 How is OTEC caused?
 a) By wind energy b) By geothermal energy
 c) By solar energy d) By gravitational force

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Constituent of biogas is/are_____.
- Q.12 Name any two energy conservation devices.
- Q.13 WECS stands for_____.
- Q.14 For working a wind mill, the minimum speed of wind shouldbe_____.
- Q.15 Full form of KVIC is_____.
- Q.16 Write any two example of non-conventional energy sources.

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- Q.17 Biogas digester.
- Q.18 SI unit of energy is_____.
- Q.19 Write any two limitations of renewable energy.
- Q.20 Enlist different types of furnaces used for biomass combustion.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define bio-gas, its composition and benefits.
- Q.22 What are the physical principles of conversion of solar energy?
- Q.23 Explain the present scenario of renewable source of energy in India.
- Q.24 Differentiate between horizontal axis wind mill and the vertical axis wind mill.
- Q.25 Explain the constructional details of flat plate box solar cooker.
- Q.26 What is wind mill? Write classifications of wind mills.
- Q.27 Write advantages and disadvantages fixed dome type biogas plant.
- Q.28 Discuss about obstacles occur for implementation of renewable energy system.
- Q.29 What are the basic criteria for site selection for setting up of biogas plant?
- Q.30 Explain the maintenance and performance of wind mill.
- Q.31 Discuss about briquetting and pelletization of biomass.
- Q.32 Enlist some methods to conserve energy in domestic sector and in agriculture sector.

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